MANEX AI x THINC!

Build the next Generation of Quality Reporting

Qualitatio: The AI Environment for Industrial Enterprises

18. & 19. April 2026

AI FOR ENTERPRISE AUTONOMY

Status Quo: Missing the foundation to provide context for AI Agents

2. PRODUCT

VALUE CHAIN

Development → Supplier → Marketing → **Production** → Customer Support → Field / Service → CX / Feedback

UPSTREAM: INTERNAL FUNCTIONS

Development / R&D
Process / Product FMEA, Reliability, Manufacturability

Supplier Quality Engineering
Qualification, Supplier Audit, Quality Metrics (ppm, DPMO)

Marketing / GTM
Social Media Campaigns, Market Analysis, Warranty

Production / Operations
MES, QMS, ERP, Quality Gates

Data silos and missing unifying architecture

DOMAINS

- Market Research
- Central Quality
- Plant Quality (indirect)
- Plant Quality (direct)
- Process Planner (Central, Plant)
- Technology Planning
- Supplier Quality (Central, Plant)
- Business Analytics
- Marketing
- ...

DATA SILOS

- CX Platform
- Warranty
- Field Service
- MES
- Mfg. Quality
- Audit Reports
- ERP
- Marketing Sys.
- CRM
- Dev. Backlog
- Req.-Sheets
- QMS / CAQ

Quality indicators are lost due to high complexity

High FTE effort for overall process

Customer voices get lost and market adaptations are very slow

DOWNSTREAM: CUSTOMER FEEDBACK

Social Media / Forums / Press
Reddit, Twitter/X, Facebook, Industry Forums, Communities, Media

Dealers / Shops / E-Com
Ratings, Return Reasons, Product Feedback, Reviews

NPS / Surveys
Customer Satisfaction, CSAT, Product Ratings

Service / Warranty
Complaints, Repairs, Service Tickets, RMA

IoT / Sensor Data
Product Telemetry, Usage Data, Predictive Maintenance

Solution: Enable AI Agents by Manex's Manufacturing Domain

2. PRODUCT

END-TO-END AI-POWERED DATA INTEGRATION ALONG THE ENTIRE VALUE CHAIN

Development → Supplier → Marketing → **Production** → Customer Support → Field / Service → CX / Feedback

UPSTREAM: INTERNAL FUNCTIONS

Development / R&D
Process / Product FMEA, Reliability, Manufacturability
AI Agents

Supplier Quality Engineering
Qualification, Supplier Audit, Quality Metrics (ppm, DPMO)
AI Agents

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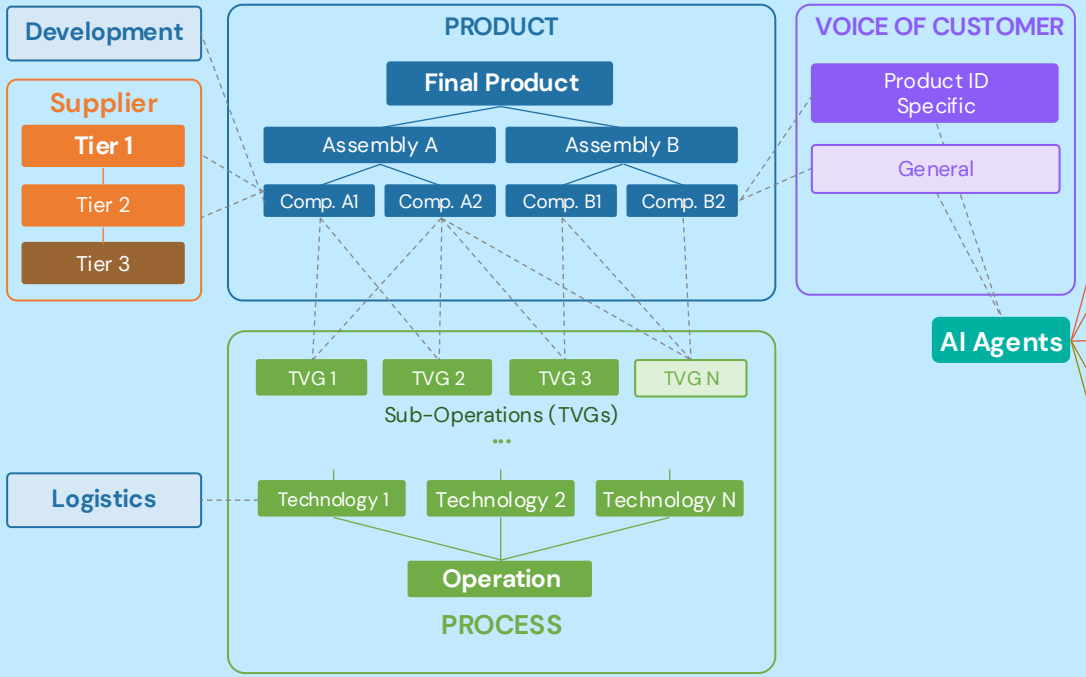
AI Agents

Production / Operations
MES, QMS, ERP, Quality Gates
AI Agents

Manex AI Manufacturing Domain (Ontology)



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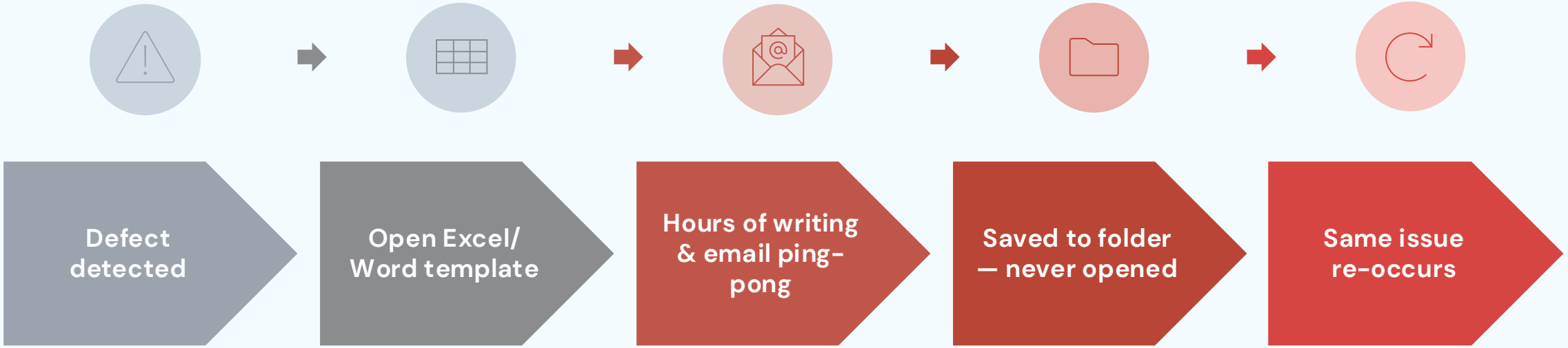
INITIATIVE Each customer voice is assigned an initiative that is semi-automatically implemented by agents in development, supplier management, marketing, production, etc.

When a quality issue hits the shopfloor, the response is manual, fragmented, and slow



This is the 8D / FMEA process at most manufacturers today.

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From static document to a living, data-connected quality workflow

TODAY

- Static Excel docs
- Disconnected from data
- No follow-up, no tracking

Hackathon Solution

TOMORROW

- **Linked to real data**
- **AI-generated insights**
- **Tracked & closed out**

The bridge from static to living — that's what you build this weekend.

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Build an intelligent Quality Resolution Module across three pillars

1

INTELLIGENT GENERATION

Use LLMs to analyze structured data — test parameters, error codes, complaint texts. Auto-generate problem descriptions and root-cause hypotheses.

2

INNOVATIVE VISUALIZATION

Rethink how failure analysis is displayed. Fault trees, timelines, Pareto charts, BOM traceability — say goodbye to boring tables.

3

CLOSED-LOOP WORKFLOW

The report must not remain static. Create initiatives, assign owners, track progress, close the loop.

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A complete demo setup — so you can focus on the solution, not on data engineering

REALISTIC DATASET
Synthetic but realistic manufacturing data across 19 tables(~7500 entries). Production, quality, supplier info, test results, complaint texts, defect images — all connected.

FOUR ROOT-CAUSE STORIES
We embedded four explicit failure patterns and documented them upfront. This is not a data treasure hunt. Your job is to think about what to do with them.

MULTIPLE ACCESS PATHS
REST API, SQL editor in the browser, or direct database connection. Pick whatever fits your stack. Connect in under 5 minutes.

WRITE-BACK CAPABILITY
Create initiatives and corrective actions back into the system. This is how you demo the closed-loop workflow.

All technical details, endpoints, schema, credentials, are in the GitHub docs and your team handout.

The challenge is not data engineering. The challenge is what you build on top of it.

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Understand the real problem first — then design a solution a quality engineer would love to use every day

PHASE 1: THINK

Understand the pain

Step into the shoes of a quality or process engineer. What is their daily frustration? What would genuinely make their work better?

Challenge the process itself

Maybe the 8D format is the problem. Maybe the whole reporting paradigm needs rethinking. Do not just digitize what exists.

Think business value and UX

Who is the end user? What does their workflow actually look like? Design for them, not for a demo jury.

PHASE 2: BUILD

Prototype your solution using our sandbox. Iterate. Demo.

The winning teams rethink the process with creative solutions, not just digitize it.

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Questions, ideas, career interests? We are here during the weekend

Simon Bestler

Product Impact Engineer (FDE)

Discord: @simon_bestler

- *Case Questions, Technical Issues*
- *FDE Recruiting*

Jan-Frederik Buelow

Tech Sales Lead

- *Manex Business Perspective, Sales*
- *Recruiting*

Book a chat with the Manex Team

About Manex, careers, or what we are building

We are your partners. Let's build something great together!

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